

NER Landslide Early Warning System - Tech Stack

Authoritative Version & File Path Reference for SIH26001 Jury

1. Frontend Dashboard & GIS Stack

Layer	Technology	Version	Path / Source	Purpose
Core Framework	Next.js App Router	14.1.3	frontend/package.json	SSR / PWA web client shell
UI Library	React / React DOM	18.2.0	frontend/package.json	Reactive state & lifecycle
Language	TypeScript	5.4.2	frontend/package.json	Strict static type safety
Styling	Tailwind CSS	3.4.1	tailwind.config.ts	Dark-slate command center theme
GIS Mapping	Leaflet	1.9.4	components/HeatmapViewer.tsx	Interactive 2D spatial overlays
Basemap	Esri Dark Gray Canvas	REST Tiles	HeatmapViewer.tsx	Watermark-free dark GIS tiles
Time-Series	Recharts	3.10.1	WeatherForecastModal.tsx	Dual-axis rain vs. risk curves
Offline Cache	Service Worker API	W3C Spec	public/sw.js	App shell & tile caching
Offline DB	IndexedDB API	Native	lib/offlineDb.ts	Local queue for citizen reports
Voice TTS	Web Speech API	Browser	components/AlertBanner.tsx	Zero-bandwidth spoken bulletins
Audio Siren	Web Audio API	Browser	EvacuationSirenModal.tsx	Real-time acoustic alert warble

2. Backend REST API & Asynchronous Architecture

Layer	Technology	Version	Location	Purpose
API Framework	FastAPI (Async)	0.110.0+	backend/requirements.txt	High-throughput asynchronous REST
ASGI Server	Uvicorn	0.28.0+	backend/requirements.txt	Async server runtime on Port 8000
Validation	Pydantic / Settings	2.6.0+	backend/app/schemas/	Strict data validation schemas
Database	PostgreSQL + PostGIS	15 / 3.3	infra/docker-compose.yml	Spatial indexed database (EPSG:4326)
ORM Layer	SQLAlchemy + GeoAlchemy2	2.0+ / 0.14+	backend/app/models/	Async spatial query abstractions
Message Broker	Redis (Alpine)	7-alpine	infra/docker-compose.yml	In-memory cache & Celery broker
Task Queue	Celery	5.3.6+	backend/app/celery_worker.py	Background risk evaluation tasks

3. Machine Learning Microservice & File Paths

Function	Model / Architecture	Module Path	Artifact / Status
Static LSI	XGBoost Classifier	ml-engine/susceptibility/susceptibility_model.py	baseline_xgb_model.joblib (Trained)
Baseline Model	Random Forest Classifier	ml-engine/baseline/train_baseline.py	baseline_rf_model.joblib (Trained)
2-6h Nowcast	2-Layer PyTorch LSTM	ml-engine/nowcast/lstm_nowcast.py	Architecture ready (Awaiting seq data)
24-48h Forecast	Deterministic Hydro-Load	ml-engine/nowcast/forecast_model.py	Algorithmic Forward Model (Done)
Road Graph	NetworkX Reachability	ml-engine/spatiotemporal/road_network_graph.py	13-node topological graph (Done)
Feature SHAP	TreeSHAP Local Attrib.	ml-engine/explainability/shap_explainer.py	Local baseline feature offset (Done)
Multilingual	Natural Language Engine	ml-engine/explainability/natural_language_explainer.py	Hindi/English text engine (Done)
Retraining	Ground-Truth Refit	ml-engine/feedback_loop/retrain_pipeline.py	Automated CSV append & refit (Done)

4. Deployment Topology

- Production Web Edge: Vercel PWA (<https://ner-early-warning-system.vercel.app>), automated git CI/CD.
- Container Topology: infra/docker-compose.yml orchestrates PostGIS, Redis, FastAPI backend, and ML microservice.
- NIC MeghRaj Cloud Ready: Kubernetes deployment manifests in infra/k8s/ with zero cloud-vendor lock-in.